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Midterm

1.

Recall that if $f(\mathbf{x})$ and $g((x))$ are probability distributions, then the Kullback-Leibler divergence of g from f is

$$D(f,g) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(\mathbf{x}) \log[\frac{f(\mathbf{x})}{g(\mathbf{x})}] d\mathbf{x}.$$

2.

The columns in the file `midterm_exam_data.txt` are the X and Y values of a function which is a sum of cosines with added Gaussian noise.

(a)

For various values of n, fit an nth order cosine series ($y = \frac{a_0}{2} + \sum_{k=1}^{n} a_k \cos kx$) to the data. What would you say is the most likely value of n?

(b)

Estimate the level of the noise.

3.

Consider a Bayesian classifier, with distributions being n-dimensional multivariate Bernoulli random variables. The conditional probability for a given category is

$$P(\mathbf{x}|\boldsymbol{\theta}) = \prod_{i=1}^{n} \theta_i^{x_i}(1-\theta_i)^{1-x_i},$$

and let $\mathcal{D} = \{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ be a set of k samples independently drawn from this probability density.

(a)

If $\mathbf{s} = (s_1, \ldots, s_n)^T$ is the sum of the k samples, show that

$$P(\mathcal{D}|\boldsymbol{\theta}) = \prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i},$$

With the setup of the problem with independent samples drawn from a Bernoulli distribution, this is simply the likelihood function. We can therefore just express this as the product of the probability densities:

$$P(\mathcal{D}|\boldsymbol{\theta}) = \mathcal{L}(\boldsymbol{\theta}|D) = \prod_{j=1}^{k} P(\mathbf{x}_j|\boldsymbol{\theta})$$

$$= \prod_{j=1}^{k}(\prod_{i=1}^{n} \theta_i^{x_{ij}}(1-\theta_i)^{1-x_{ij}}) = \prod_{j=1}^{k}\prod_{i=1}^{n} \theta_i^{x_{ij}}(1-\theta_i)^{1-x_{ij}}$$

$$= \prod_{i=1}^{n}\prod_{j=1}^{k} \theta_i^{x_{ij}}(1-\theta_i)^{1-x_{ij}}$$

$$= \prod_{i=1}^{n} \theta_i^{\sum_{j=1}^{k} x_{ij}}(1-\theta_i)^{\sum_{j=1}^{k} 1-x_{ij}}$$

Substituting the sum s in accordingly as given gives us exactly the desired result:

$$\boxed{P(\mathcal{D}|\boldsymbol{\theta}) = \prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}$$

(b)

Assuming a uniform prior distribution for $\boldsymbol{\theta}$ show that

$$P(\boldsymbol{\theta}|\mathcal{D}) = \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!}\theta_i^{s_i}(1-\theta_i)^{k-s_i},$$

From part (a), we can find this using Bayes' thoerem:

$$P(\boldsymbol{\theta}|\mathcal{D}) = \frac{P(\mathcal{D}|\boldsymbol{\theta}) \cdot P(\boldsymbol{\theta})}{P(\mathcal{D})}$$

$$= \frac{P(\mathcal{D}|\boldsymbol{\theta}) \cdot P(\boldsymbol{\theta})}{\int P(\mathcal{D}|\boldsymbol{\theta}) \cdot P(\boldsymbol{\theta})}$$

Substituting in our result from part (a):

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i} \cdot P(\boldsymbol{\theta})}{\int \prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i} \cdot P(\boldsymbol{\theta})}$$

And using the assumption of the uniform distribution for $\boldsymbol{\theta}$, i.e. $P(\boldsymbol{\theta}) = 1$ for $0 \leq \theta_i \leq 1$:

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i} \cdot 1}{\int \prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i} \cdot 1}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \int_0^1 \theta_i^{s_i}(1-\theta_i)^{k-s_i} d\theta_i}$$

We use integration by parts in the denominator:

- $u = (1-\theta_i)^{k-s_i}$
- $du = (1-\theta_i)^{k-s_i-1} \cdot (k-s_i) \cdot -1 d\theta_i = (s_i-k)(1-\theta_i)^{k-s_i-1} d\theta_i$
- $v = \frac{\theta_i^{s_i+1}}{s_i+1}$
- $dv = \theta_i^{si} d\theta_i$ And substituting accordingly:

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n}(1-\theta_i)^{k-s_i} \cdot \frac{\theta_i^{s_i+1}}{s_i+1}|_0^1 - \int_0^1 \frac{\theta_i^{s_i+1}}{s_i+1} \cdot (s_i-k)(1-\theta_i)^{k-s_i-1} d\theta_i}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} 0 - 0 - \int_0^1 \frac{\theta_i^{s_i+1}}{s_i+1} \cdot (s_i-k)(1-\theta_i)^{k-s_i-1} d\theta_i}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} 0 - 0 - \int_0^1 \frac{\theta_i^{s_i+1}}{s_i+1} \cdot (s_i-k)(1-\theta_i)^{k-s_i-1} d\theta_i}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} -\frac{s_i-k}{s_i+1} \int_0^1 \theta_i^{s_i+1}(1-\theta_i)^{k-s_i-1} d\theta_i}$$

We can see that from repeatedly applying integration by parts, we eventually reduce the exponent of the $1-\theta_i$ term to 0, while the exponent of the θ_i term increases until k, giving us:

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)(k-s_i-1)\ldots 1}{(s_i+1)(s_i+2)\ldots k} \int_0^1 \theta_i^{k}(1-\theta_i)^{0} d\theta_i}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)(k-s_i-1)\ldots 1}{(s_i+1)(s_i+2)\ldots k} \int_0^1 \theta_i^{k}(1-\theta_i)^{0} d\theta_i}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)(k-s_i-1)...1}{(s_i+1)(s_i+2)...k} \int_0^1 \theta_i^k d\theta_i}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)(k-s_i-1)...1}{(s_i+1)(s_i+2)...k} [\frac{\theta_i^{k+1}}{k+1}]_0^1}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)(k-s_i-1)...1}{(s_i+1)(s_i+2)...k} (\frac{1}{k+1} - 0)}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)!}{(s_i+1)(s_i+2)...k+1}}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)!}{(s_i+1)(s_i+2)...k+1} \cdot \frac{s_i!}{s_i!}}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)!s_i!}{(k+1)!}}$$

$$= \prod_{i=1}^{n} \frac{\theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\frac{(k-s_i)!s_i!}{(k+1)!}}$$

$$\boxed{P(\boldsymbol{\theta}|\mathcal{D}) = \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}$$

exactly as we sought to find.

(c)

Plot this density for the case $n = 1, k = 1$ and for the two resulting possibilities for s_i.

With $n = 1$, we have one sample, so our only cases are either $s = 0$ or $s = 1$:

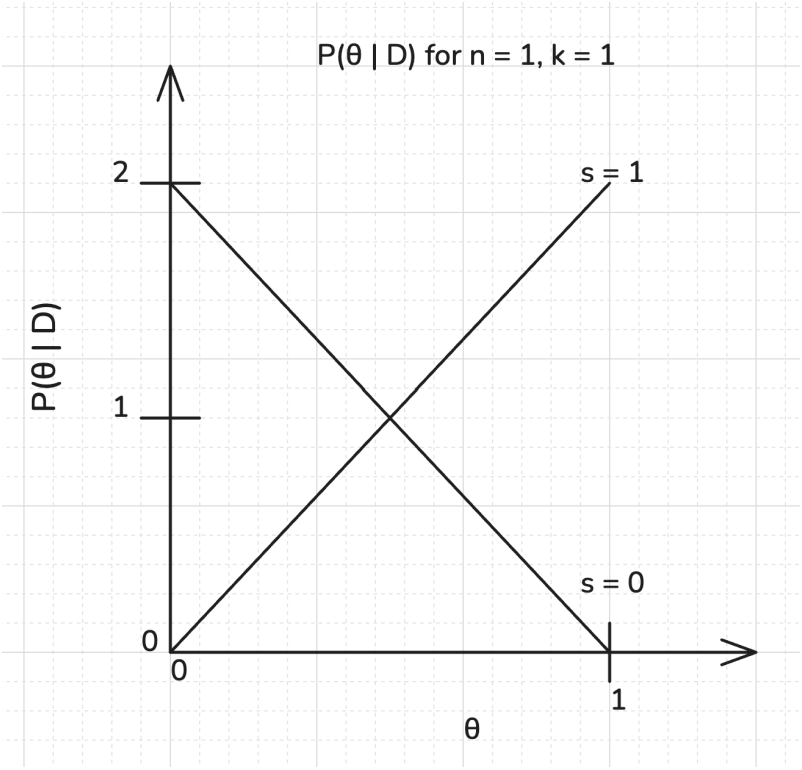
- If $s = 0$, we have

$$P(\boldsymbol{\theta}|\mathcal{D}) = \frac{(1+1)!}{0!(1-0)!} \cdot \theta^0 \cdot (1-\theta)^{1-0} = \frac{2!}{0!1!} \cdot 1 \cdot (1-\theta)^1 = 2 - 2\theta$$

- If $s = 1$, we have

$$P(\boldsymbol{\theta}|\mathcal{D}) = \frac{(1+1)!}{1!(1-1)!} \cdot \theta^1 \cdot (1-\theta)^{1-1} = \frac{2!}{1!0!} \cdot \theta \cdot (1-\theta)^0 = 2\theta$$

We plot this below:



(d)

Integrate the product $P(\mathbf{x}|\boldsymbol{\theta})P(\boldsymbol{\theta}|\mathcal{D})$ over $\boldsymbol{\theta}$ to obtain

$$P(\mathbf{x}|\mathcal{D}) = \prod_{i=1}^{n}(\frac{s_i+1}{k+2})^{x_i}(1-\frac{s_i+1}{k+1})^{1-x_i}$$

As directed, we start with

$$P(\mathbf{x}|\mathcal{D}) = \int_{\boldsymbol{\theta}} P(\mathbf{x}|\boldsymbol{\theta})P(\boldsymbol{\theta}|\mathcal{D})d\boldsymbol{\theta}$$

Substituting the given conditional probability and our finding from part (b):

$$= \int_{\boldsymbol{\theta}}(\prod_{i=1}^{n}\theta_i^{x_i}(1-\theta_i)^{1-x_i}) \cdot (\prod_{i=1}^{n}\frac{(k+1)!}{s_i!(k-s_i)!}\theta_i^{s_i}(1-\theta_i)^{k-s_i})d\boldsymbol{\theta}$$

$$= \prod_{i=1}^{n}\int_{\boldsymbol{\theta}}\theta_i^{x_i}(1-\theta_i)^{1-x_i}\frac{(k+1)!}{s_i!(k-s_i)!}\theta_i^{s_i}(1-\theta_i)^{k-s_i}d\boldsymbol{\theta}$$

$$= \prod_{i=1}^{n}\frac{(k+1)!}{s_i!(k-s_i)!}\int_{\boldsymbol{\theta}}\theta_i^{x_i}(1-\theta_i)^{1-x_i}\theta_i^{s_i}(1-\theta_i)^{k-s_i}d\boldsymbol{\theta}$$

$$= \prod_{i=1}^{n}\frac{(k+1)!}{s_i!(k-s_i)!}\int_{\boldsymbol{\theta}}\theta_i^{x_i+s_i}(1-\theta_i)^{k-s_i-x_i+1}d\boldsymbol{\theta}$$

This leads us into a very similar situation as we saw in part (b) with the repeated integration by parts. In part (b), we have exponents of s_i and $k - s_i$; here, we have exponents of $s_i + x_i$ and $k - s_i - x_i + 1$, so the $1 - \theta_i$ term now has 1 additional power relative to the θ_i term as it did in part (b). We just integrate by parts $k - s_i - x_i + 1$ times, giving us

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!}{(s_i+x_i+1)(x_i+s_i+2)\dots(k+1)} \int_0^1 \theta_i^{k+1}(1-\theta_i)^0 d\boldsymbol{\theta}$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!}{(s_i+x_i+1)(x_i+s_i+2)\dots(k+1)} [\frac{\theta_i^{k+2}}{k+2}]_0^1$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!}{(s_i+x_i+1)(x_i+s_i+2)\dots(k+1)} \cdot (\frac{1}{k+2} - 0)$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!}{(s_i+x_i+1)(x_i+s_i+2)\dots(k+2)}$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!}{(s_i+x_i+1)(x_i+s_i+2)\dots(k+2)} \cdot \frac{(s_i+x_i)!}{(s_i+x_i)!}$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!(s_i+x_i)!}{(k+2)!}$$

$$= \prod_{i=1}^{n} \frac{(k-s_i-x_i+1)!(s_i+x_i)!}{s_i!(k-s_i)!(k+2)}$$

$$= \prod_{i=1}^{n} \frac{(k-s_i-x_i+1)!(s_i+x_i)!}{s_i!(k-s_i)!(k+2)}$$

From this point, we can take a hint from the structure of the answer to see that we should be looking at the two different cases for x_i, that is, because it is Bernoulli, it must either be 1 or 0. Looking at these cases:

- If $x_i = 0$, we get

$$\frac{(k-s_i-0+1)!(s_i+0)!}{s_i!(k-s_i)!(k+2)} = \frac{(k-s_i+1)!s_i!}{s_i!(k-s_i)!(k+2)} = \frac{k-s_i+1}{k+2}$$

- If $x_i = 1$, we get

$$\frac{(k-s_i-1+1)!(s_i+1)!}{s_i!(k-s_i)!(k+2)} = \frac{s_i+1}{k+2}$$

And we then can express this using the exponents to account for the 0 / 1 on/off nature of the results. We therefore get

$$= \prod_{i=1}^{n} (\frac{k - s_i + 1}{k+2})^{1-x_i} (\frac{s_i + 1}{k+2})^{x_i}$$

$$= \prod_{i=1}^{n} (\frac{s_i + 1}{k+2})^{x_i} (\frac{k - s_i + 1}{k+2})^{1-x_i}$$

$$= \prod_{i=1}^{n} (\frac{s_i + 1}{k+2})^{x_i} (\frac{k + 2 - s_i - 1}{k+2})^{1-x_i}$$

$$\boxed{P(\mathbf{x}|\mathcal{D}) = \prod_{i=1}^{n} (\frac{s_i + 1}{k+2})^{x_i} (1 - \frac{s_i + 1}{k+2})^{1-x_i}}$$

exactly as we sought to find.

(e)

Compare the expressions for $P(\mathbf{x}|\mathcal{D})$ and $P(\mathbf{x}|\boldsymbol{\theta})$. What is $\hat{\boldsymbol{\theta}}$, the effective Bayesian estimate for $\boldsymbol{\theta}$?

To recap, we have

$$P(\mathbf{x}|\mathcal{D}) = \prod_{i=1}^{n} (\frac{s_i + 1}{k+2})^{x_i} (1 - \frac{s_i + 1}{k+2})^{1-x_i}$$

and

$$P(\mathbf{x}|\boldsymbol{\theta}) = \prod_{i=1}^{n} \theta_i^{x_i} (1 - \theta_i)^{1-x_i}$$

We see that that θ_i directly maps to $\frac{s_i+1}{k+2}$ for both the θ_i and $1 - \theta_i$ terms, which indicates to us that $\hat{\boldsymbol{\theta}}$ is $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_n)^T$ where $\hat{\theta}_i = \frac{s_i+1}{k+2}$.